

Lecture 1 Planned Notes

Sunday, May 11, 2025 9:29 PM

Review:

Curve C parametrized $\vec{r}(t)$, $a \leq t \leq b$.

has

$$\text{arc-length } l = \int_a^b \sqrt{\vec{r}' \cdot \vec{r}'} dt$$

and can be reparametrized: $\vec{r}(s)$, $0 \leq s \leq l$

$$\text{with } s = \int_a^t \sqrt{\vec{r}' \cdot \vec{r}'} d\tilde{t}$$

such that the arc-length parametrization is unit speed: $\|\vec{r}'(s)\| = 1$

Example: Catenary: $\vec{r} = [t, \cosh(t)]$, $0 \leq t \leq 1$

reparametrized $\vec{r} = [\sinh^{-1}(s), \sqrt{s^2+1}]$, $s = \sinh(t)$, $0 \leq s \leq \sinh(1)$

Note: $\vec{r}'(t) = \begin{bmatrix} 1 \\ \sinh(t) \end{bmatrix}$. so at $t = \sinh^{-1}(\frac{1}{2}) \approx 0.4812\dots$, $\vec{r}'(\sinh^{-1}(\frac{1}{2})) = \begin{bmatrix} 1 \\ \sinh(\sinh^{-1}(\frac{1}{2})) \end{bmatrix} = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$\text{so speed} = \sqrt{1^2 + (\frac{1}{2})^2} = \frac{\sqrt{5}}{2} \text{ (not unit speed)}$$

but $\vec{r}'(s) = \begin{bmatrix} \frac{1}{\sqrt{s^2+1}} \\ \frac{s}{\sqrt{s^2+1}} \end{bmatrix}$, and $s = \sinh(t)$, so at $t = \sinh^{-1}(\frac{1}{2})$, $s = \frac{1}{2}$

$$\text{Thus: } \vec{r}'(s) = \begin{bmatrix} \frac{1}{(\frac{1}{2})^2+1} \\ \frac{\frac{1}{2}}{\sqrt{(\frac{1}{2})^2+1}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{1/2}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and speed} = 1$$

So $\vec{r}'(t)$ and $\vec{r}'(s)$ are in the same direction!

$$\text{However: } \vec{r}''(t) = \begin{bmatrix} 0 \\ \cosh(t) \end{bmatrix}, \text{ while } \vec{r}''(s) = \begin{bmatrix} \frac{-s}{(s^2+1)^{3/2}} \\ \frac{-s^2}{(s^2+1)^{3/2}} + \frac{1}{(s^2+1)^{3/2}} \end{bmatrix}$$

then at $s = \sinh(t) = \frac{1}{2}$, since $\cosh(t) = \sqrt{1 + \sinh^2(t)}$, we have

$$\begin{aligned} \vec{r}''(\sinh^{-1}(\frac{1}{2})) &= \begin{bmatrix} 0 \\ \frac{1}{\sqrt{5/4}} \end{bmatrix}, \text{ but } \vec{r}''(\frac{1}{2}) = \begin{bmatrix} \frac{-1/2}{(5/4)^{3/2}} \\ \frac{-(1/2)^2}{(5/4)^{3/2}} + \frac{1}{(5/4)^{3/2}} \end{bmatrix} = \begin{bmatrix} \frac{-\sqrt{2}}{5\sqrt{5}} \\ \frac{-1/4}{(5/4)^{3/2}} + \frac{(5/4)}{(5/4)^{3/2}} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{\sqrt{2}}{5\sqrt{5}} \\ \frac{2\sqrt{2}}{5\sqrt{5}} \end{bmatrix} \end{aligned}$$

(1) $\vec{r}''(t)$ and $\vec{r}''(s)$ do not share the same direction!

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$$(2) \quad \vec{r}''(\tfrac{1}{2}) \cdot \vec{r}'(\tfrac{1}{2}) = \frac{\sqrt{2}}{5\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \left(\frac{\sqrt{2}}{5\sqrt{5} \times \sqrt{5}} \right) ((-1)(2) + (2)(1)) \\ = \frac{2}{25} (0) = 0$$

So $\vec{r}''(s)$ and $\vec{r}'(s)$ are perpendicular!